

1 Experiment Details

1.1 Participants

A total of 171 individuals participated in this study. The first 13 participants were excluded from analysis due to a data recording error, leaving a final sample of $N = 146$ participants (112 males, 34 females; $M_{\text{age}} = 22.0$ years, $SD = 1.9$, range: 19–28 years). All participants reported normal or corrected-to-normal vision (68.5% normal, 31.5% corrected), and the majority were right-handed (95.2%). Participants were randomly assigned to one of two between-subjects conditions: the Natural Boundary (NB) condition ($n = 79$) or the Abrupt Boundary (AB) condition ($n = 67$).

1.2 Stimuli

Forty YouTube Shorts were selected as stimulus videos. To identify event boundaries in each clip, an independent group of annotators watched the videos and marked coarse-grained event boundaries via keypress. Consensus boundaries were derived by identifying temporal locations where multiple annotators agreed.

1.2.1 Condition Manipulation

Using the consensus boundaries, two versions of each video were created to manipulate event boundary continuity. In the **Natural Cut (NB)** condition, videos played continuously to their natural conclusion without interruption. Event boundaries occurred exactly when viewers would naturally expect them, preserving cognitive continuity and maintaining the integrity of the event structure. In the **Abrupt Cut (AB)** condition, videos were terminated *before* a consensus boundary and immediately resumed at the onset of the subsequent event. Video duration was matched across conditions to ensure that temporal length did not serve as a confounding factor.

1.2.2 Recognition Test Stimuli

For the recognition memory test, two types of target frames were extracted from each video. **Before-Boundary (BB)** frames were still images captured just prior to an event boundary, representing the temporally critical region where event transitions occur. **Event-Middle (EM)** frames were captured from the centre of an event, temporally distant from any boundary. For each target frame, a corresponding lure frame was selected.

1.3 Design

The experiment employed a 2×2 mixed design. The between-subjects factor was Boundary Condition (Natural Cut vs. Abrupt Cut), and the within-subjects factor was Frame Type (Before-Boundary vs. Event-Middle).

1.4 Procedure

1.4.1 Phase 1: Encoding

Participants were instructed to watch a series of short video clips attentively in preparation for a subsequent memory test. To ensure attentive viewing, five videos were repeated during the encoding phase as vigilance checks. Participants who exceeded 27.05 minutes during the encoding phase were flagged as inattentive and excluded.

1.4.2 Phase 2: Recognition Memory Test

Following encoding, participants completed a two-alternative choice. On each trial, two still frames from the same video were presented side-by-side: one target frame and one lure frame. Participants indicated which frame was the target via keypress (left or right). Response accuracy (`resp.corr`: 0 = incorrect, 1 = correct) and response time (`resp.rt`: seconds from stimulus onset to keypress) were recorded. Immediately following each recognition decision, participants rated their confidence on a 5-point Likert scale (1 = *very unconfident*, 5 = *very confident*), recorded as `conf_radio.response`. Each participant completed approximately 40 recognition trials.

1.5 Dependent Measures

Three primary dependent variables were analysed. **Recognition accuracy** (`resp.corr`) was a binary measure of correct target identification, aggregated as proportion correct per participant per condition. **Response time** (`resp.rt`) indexed retrieval speed and decision difficulty. **Confidence** (`conf_radio.response`) provided a measure of metacognitive monitoring and subjective memory strength. In addition, two composite indices were computed. The **Recognition Memory Index (REC)** measured bias-corrected discrimination between targets and foils, calculated as $P(\text{"old"} \mid \text{Target}) - P(\text{"old"} \mid \text{Foil})$. The **Lure Discrimination Index (LDI)** measured high-fidelity mnemonic discrimination, calculated as $P(\text{"similar"} \mid \text{Lure}) - P(\text{"similar"} \mid \text{Foil})$. Recognition performance was analysed separately for BB and EM frames to test predictions concerning boundary-specific encoding effects.

2 Literature

The following works collectively motivate the experimental design and generate the predictions tested in this study.

Event Segmentation Theory (EST) posits that the perceptual system continuously parses the ongoing stream of experience into discrete, meaningful units referred to as events. Central to the theory is the notion of an *event model*: a working memory representation that guides perceptual processing by generating predictions about what will happen next. This model remains stable as long as incoming perceptual information is consistent with current predictions. However, when prediction error increases when the environment becomes less predictable the current event model can no longer adequately account for new input, and a model update is triggered [1].

Building directly on EST, [2] proposed the Event Horizon Model, and the *boundary advantage*. This principle holds that event boundaries serve as privileged anchor points in LTM, such that the processing occurring at boundaries which involves updating and consolidating the prior event model is especially conducive to durable memory encoding. Recall and recognition of information presented at or near event boundaries tends to be superior to information from event interiors. [2] also highlight evidence that individuals who show less agreement with others when segmenting events perform more poorly on subsequent memory tests, suggesting that reliable event segmentation is mechanistically linked to effective encoding. A closely related finding reported by schwab2004 demonstrated that deleting temporal segments from within events impairs memory less than deleting equivalent segments at event boundaries, directly demonstrating that boundaries represent the most cognitively critical moments in a continuous experience. Empirical support for the link between event boundaries and memory encoding in film contexts was provided early by [4]. In this study, participants viewed a feature film

into which advertisement breaks were inserted at either natural event boundaries or at non-boundary locations within events. Advertisements inserted at natural boundaries were associated with improved subsequent recall of the surrounding film content, whereas advertisements inserted at non-boundary locations impaired recall.

The specific prediction that items presented near event boundaries are better encoded than items from event interiors was tested directly by [5]. Across three experiments, participants watched films depicting everyday activities while performing an incidental encoding task involving object recognition. The key manipulation was whether objects appeared at moments coinciding with an event boundary or during stable periods within an event. Across all experiments, objects that were present at the time of boundary perception were subsequently recognised with higher accuracy than those presented within event interiors. Notably, this boundary advantage was observed even across the boundary that is, objects from the preceding event were still better remembered when a boundary followed shortly after their presentation. This finding maps directly onto the design of the present study. The BB frames are drawn from the temporal region immediately preceding an event boundary, the zone in which [5] observed the strongest encoding advantage. The EM frames, by contrast, are drawn from the event interior, where encoding advantage is minimal. The NB vs. AB manipulation then allows us to test whether the boundary advantage for BB frames depends on the natural boundary transition completing undisturbed, or whether it persists even when the boundary moment is artificially removed.

3 Descriptive Statistics

3.1 Sample Characteristics

The final analytical sample consisted of 146 participants who completed a total of 6,800 recognition memory trials, averaging approximately 46.6 trials per participant. Participants were distributed across two experimental conditions: 79 participants (54.1%) in the Natural Boundary (NB) condition and 67 participants (45.9%) in the Abrupt Boundary (AB) condition. Demographic characteristics are summarised in Table 1.

Table 1: Sample Demographic Characteristics by Condition

Characteristic	NB ($n = 79$)	AB ($n = 67$)	Total ($N = 146$)
Age: M (SD)	22.2 (2.1)	21.9 (1.6)	22.0 (1.9)
Male / Female	63 / 16	49 / 18	112 / 34
Right-handed	75 (94.9%)	64 (95.5%)	139 (95.2%)
Normal vision	54 (68.4%)	46 (68.7%)	100 (68.5%)
Corrected-to-normal vision	25 (31.6%)	21 (31.3%)	46 (31.5%)

3.2 Recognition Accuracy

Overall recognition accuracy was high ($M = 0.857$, $SD = 0.077$), indicating that participants correctly identified the target frame in approximately 86% of trials. The AB condition also exhibited greater performance variability and a lower minimum accuracy score, suggesting that boundary disruption produced less consistent encoding across participants. Descriptive statistics for accuracy by condition are presented in Table 2.

Table 2: Recognition Accuracy by Condition

Condition	M	SD	Mdn	Min	Max	IQR
Natural Boundary (NB)	0.872	0.069	0.875	0.675	0.988	0.825–0.925
Abrupt Boundary (AB)	0.838	0.084	0.850	0.575	0.975	0.800–0.900
Overall	0.857	0.077	0.875	0.575	0.988	0.825–0.900

3.3 Response Time

Mean response time across all participants was 5.718 seconds ($Mdn = 5.470$ s, IQR: 4.740–6.158 s), ranging from 3.251 to 12.180 seconds. Contrary to a simple speed-accuracy tradeoff, NB participants took slightly longer to respond yet achieved higher accuracy compared to AB participants, who responded faster but less accurately. This pattern suggests that preserved event boundaries facilitate more deliberate retrieval of richer memory traces, whereas disrupted boundaries may prompt faster but less accurate decisions. At the trial level, both conditions showed substantially wider RT distributions, with the AB condition containing an extreme outlier at 71.085 seconds. Participant-level and trial-level response time statistics are presented in Tables 3 and 4, respectively.

Table 3: Response Time (seconds) by Condition — Participant Level

Condition	M	Mdn	Min	Max	IQR
Natural Boundary (NB)	5.834	5.534	3.256	12.180	4.904–6.318
Abrupt Boundary (AB)	5.582	5.399	3.251	11.633	4.665–5.954
Overall	5.718	5.470	3.251	12.180	4.740–6.158

Table 4: Response Time (seconds) by Condition — Trial Level

Condition	M	Mdn	Min	Max	IQR
Natural Boundary (NB)	5.802	4.731	0.276	54.723	3.433–6.844
Abrupt Boundary (AB)	5.572	4.513	1.230	71.085	3.335–6.570
Overall	5.693	4.626	0.276	71.085	3.380–6.699

3.4 Confidence Ratings

Participants reported relatively high confidence overall ($M = 4.149$, $SD = 0.422$, $Mdn = 4.225$), corresponding to approximately 83% of the maximum possible rating. The pattern of confidence differences across conditions mirrored the accuracy findings: NB participants reported higher mean confidence than AB participants, and the AB condition showed marginally greater variability. The alignment between objective accuracy and subjective confidence indicates adequate metacognitive awareness, with participants able to monitor the quality of their own memory retrieval. Confidence statistics are presented in Table 5.

Table 5: Confidence Ratings (1–5 scale) by Condition

Condition	M	SD	Mdn	Min	Max	IQR
Natural Boundary (NB)	4.209	0.407	4.275	3.000	5.000	3.888–4.488
Abrupt Boundary (AB)	4.077	0.431	4.100	2.900	4.900	3.725–4.313
Overall	4.149	0.422	4.225	2.900	5.000	3.825–4.425

3.5 Trial-Level Performance

At the trial level, 5,822 of the 6,800 trials (85.6%) resulted in correct target identification. When broken down by condition, NB trials yielded higher accuracy than AB trials, corroborating the participant-level findings. Response choice distribution was near-equal across the full sample, indicating no systematic spatial response bias. Trial-level accuracy by condition is summarised in Table 6.

Table 6: Trial-Level Accuracy by Condition

Condition	Total Trials	Correct	Incorrect	Accuracy
Natural Boundary (NB)	3,560	3,102	458	87.1%
Abrupt Boundary (AB)	3,240	2,720	520	84.0%
Total	6,800	5,822	978	85.6%

3.6 Summary

In summary, three consistent patterns emerge from the descriptive statistics. First, NB participants demonstrated superior recognition memory performance across all measures—higher accuracy (87.2% vs. 83.8%), higher confidence (4.21 vs. 4.08), and lower performance variability—providing preliminary support for the boundary-specific encoding hypothesis. Second, the AB condition was associated with faster yet less accurate responses, suggesting differences in retrieval strategy rather than a simple speed-accuracy tradeoff. Third, the correspondence between accuracy and confidence across conditions indicates preserved metacognitive monitoring, implying that the experimental manipulation affected memory strength without impairing participants’ ability to evaluate the quality of their own memory judgments.

4 Data Preprocessing

Key response variables including accuracy (`resp.corr`), response time (`resp.rt`), and confidence ratings (`conf_radio.response`) were converted to numeric format to ensure computational consistency. We adopted *no-data-loss* approach for participant retention. All 146 participants who completed the recognition memory task were retained in the final sample, regardless of demographic data completeness. Among these participants, 38 individuals (26.0%) had missing demographic information in one or more variables. For the continuous variable age, missing values were imputed using the median value within each experimental condition (Natural Boundary vs. Abrupt Boundary). For categorical variables (gender, handedness, vision), missing values were replaced with the mode within each respective condition.

5 Data Visualisations

Four distributional visualisations were produced to characterise the sample and primary outcome variables. Age distribution was examined using a histogram with 20 equal-width bins at the participant level ($N = 146$), with mean age overlaid as a vertical reference line to assess sample homogeneity and identify potential age-related confounds. Confidence ratings were visualised as a bar chart rather than a histogram, as `conf_radio.response` is an ordinal scale (1–5) and discrete bars better respect the categorical nature of the data, preventing misleading interpolation between rating levels; frequency counts were aggregated across all 6,800 trials and displayed above each bar. Recognition accuracy was examined at the participant level using a histogram with 25 equal-width bins, where each participant’s accuracy was computed as the proportion of correct responses across their trials, and mean accuracy was overlaid as a reference line to assess distributional shape and distance from chance-level performance (0.5). Response time was characterised at the trial level ($N = 6800$) using a 60-bin histogram with both mean and median overlaid, as the separation between these two measures of central tendency directly indicates the degree of positive skew inherent in RT distributions and informs the choice between parametric and non-parametric inferential tests.

Violin plots were used to compare distributions across conditions (AB vs. NB) and frame types (BB vs. EM) on response time, confidence, and accuracy. Each violin combines kernel density estimation with embedded box plot statistics, making distributional shape, spread, and central tendency simultaneously visible — an advantage over standard box plots when examining whether experimental manipulations shift the full distribution rather than just the mean. A final panel plots confidence stratified by response

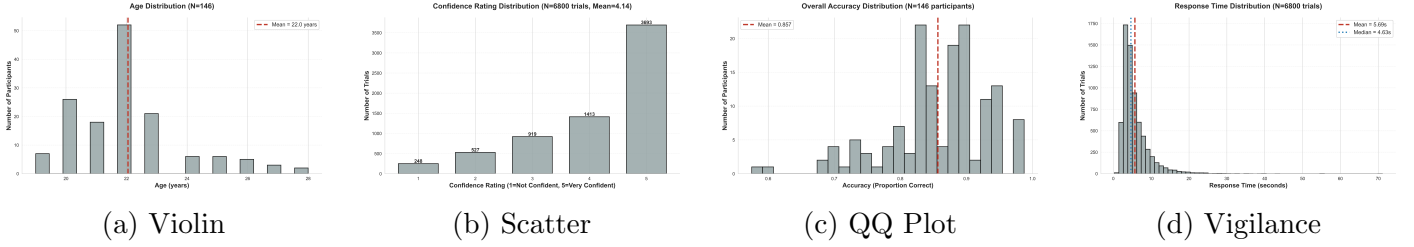


Figure 1: Visualization across different analyses

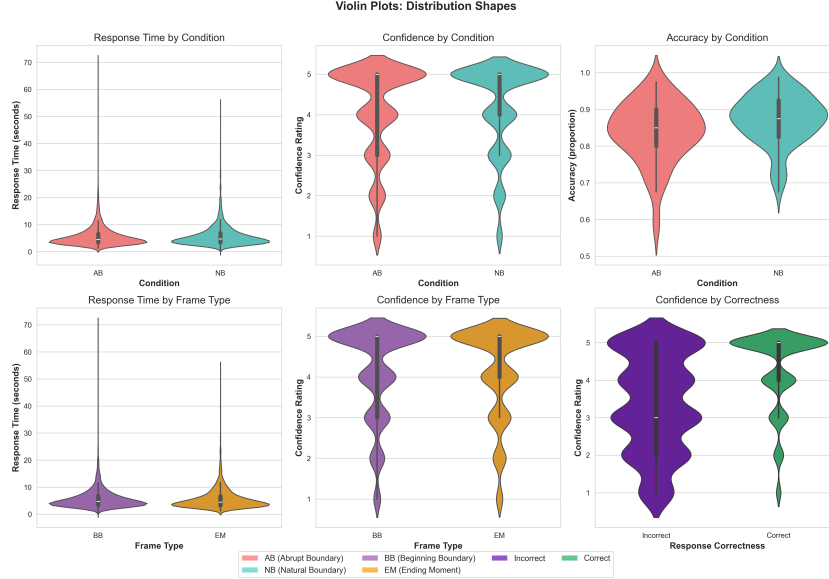


Figure 2: Violin plots across different conditions

correctness, serving as a metacognitive calibration check: well-calibrated participants should show clearly higher confidence on correct than incorrect trials.

5.1 Normality Assessment

Normality was assessed for four variables using Q-Q plots alongside three formal tests: the Shapiro-Wilk test, the Kolmogorov-Smirnov (KS) test, and the D’Agostino-Pearson test. Q-Q plots compare empirical quantiles against theoretical quantiles from a standard normal distribution $N(0, 1)$; points lying on the diagonal reference line indicate normality, while systematic deviations indicate skewness, heavy tails, or outliers. The Shapiro-Wilk statistic W measures the correlation between observed order statistics and expected normal order statistics, with values near 1 indicating normality and rejection of H_0 at $p < .05$ indicating departure. The KS statistic $D = \max_x |F_n(x) - F(x)|$ quantifies the maximum distance between the empirical and theoretical cumulative distribution functions. The D’Agostino-Pearson test combines skewness and kurtosis into a single K^2 statistic distributed as χ^2 with 2 degrees of freedom under H_0 .

All four variables departed significantly from normality. Trial-level response time was non-normal across all three tests ($D = 0.154$, $p < .001$; $K^2 = 5815.51$, $p < .001$), consistent with the strong positive skew typical of RT distributions. Participant-level mean RT was similarly non-normal ($W = 0.871$, $p < .001$; $D = 0.155$, $p = .002$; $K^2 = 54.89$, $p < .001$). Participant-level recognition accuracy also violated normality ($W = 0.943$, $p < .001$; $D = 0.122$, $p = .024$; $K^2 = 21.12$, $p < .001$), reflecting a negatively skewed distribution with performance concentrated near ceiling. Mean confidence per participant showed borderline non-normality on the Shapiro-Wilk test ($W = 0.979$, $p = .025$) but did not reach significance on the KS test ($D = 0.086$, $p = .223$) or D’Agostino-Pearson test ($K^2 = 3.69$, $p = .158$), suggesting approximate normality at the participant level. Given these results, response time analyses employed log-transformed values, and non-parametric tests were considered as alternatives where distributional assumptions were critically violated. Vigilance performance was analysed to

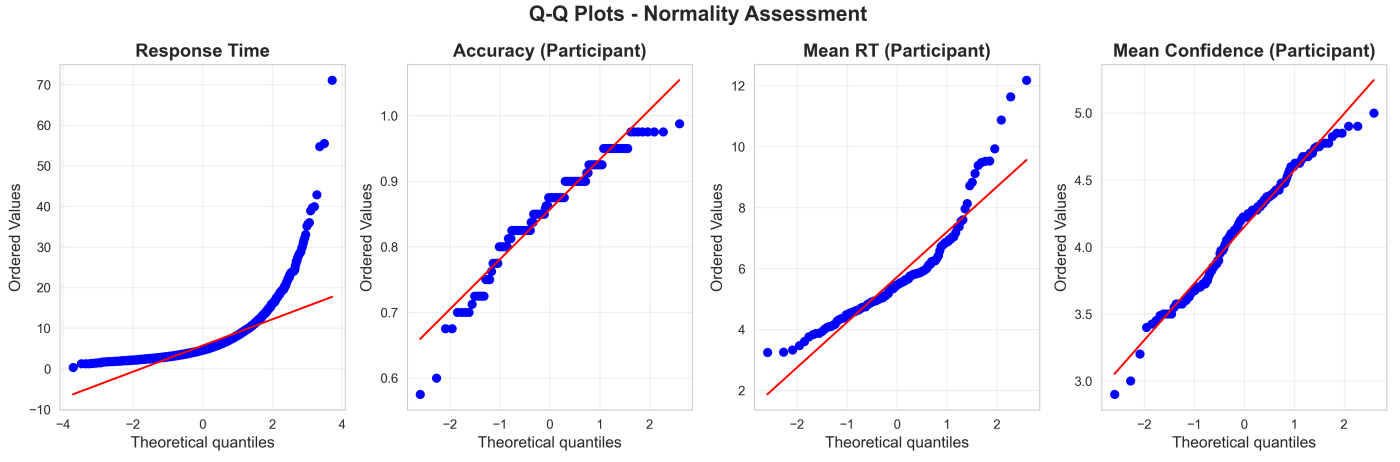


Figure 3: QQ plots



Figure 4: Vigilance Check Performance

verify attentiveness during encoding and examine its relationship to recognition memory. Three panels were produced: a pie chart showing the overall pass/fail distribution, a grouped bar chart comparing pass rates between AB and NB conditions tested via a chi-square test of independence, and a violin plot comparing recognition accuracy between vigilance-pass and vigilance-fail participants tested via an independent-samples t -test or Mann-Whitney U test depending on normality. Vigilance status was extracted from the encoding phase rows of the raw CSV files, where a participant was classified as passing if they correctly pressed the spacebar upon detecting a repeated video. Of the 161 participants with usable encoding data, 159 (98.8%) passed the vigilance check, indicating that participants were generally attentive during encoding and that the repeated-video detection task was appropriately calibrated.

6 Research Questions

7 Future works

References

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8 Team details